Saugat Aryal, M.S., E.I.T.

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PROFESSIONAL SUMMARY

Civil engineer, E.I.T., and Ph.D. candidate specializing in hydrologic and hydraulic modeling, flood-risk assessment, and water-resources engineering. Combines large-scale river and floodplain simulations, satellite-informed model evaluation, and Python/GIS analysis with professional experience in roadway drainage and culvert design.

LICENSURE & EXAMINATIONS

- **PE Civil: Water Resources and Environmental Exam – Passed** Oct 2026
- **Engineer-in-Training (E.I.T.)** – FE Civil Exam Passed Nov 2024

EDUCATION

Michigan State University (MSU), Civil Engineering East Lansing, MI
Ph.D. in Civil Engineering (Expected Mar 2027) Aug 2023 – Present
M.S. in Civil Engineering (GPA: 3.892/4.0) Dec 2025
Tribhuvan University (IOE, Pulchowk Campus) Kathmandu, Nepal
B.E. in Civil Engineering (GPA: 3.9) Nov 2017 – Jul 2022

TECHNICAL SKILLS

Hydrology/Hydraulics HEC-RAS (1D), HEC-HMS, SRH-2D/SMS, CaMa-Flood, SWMM, SWAT, MODFLOW
Engineering Design/GIS Civil 3D, AutoCAD, ArcGIS, QGIS
Programming/Analysis Python, R

PROFESSIONAL EXPERIENCE

Graduate Research Assistant – Multi-scale FEW Lab, MSU Aug 2023 – Present

- Modeled river and floodplain dynamics using **CaMa-Flood** across 11 Asian basins and 32 climate–hydrology model combinations to evaluate discharge, flood extremes, and water availability
- Built **Python/GIS** workflows for ensemble processing, spatial analysis, and flood mapping to evaluate uncertainty and hydrologic change
- Evaluated a 15-member river assimilation ensemble with synthetic SWOT-like water-level observations; improved water-level accuracy while assessing discharge stability

Civil Engineer – PS Nirman Sewa Jan 2023 – Aug 2023

- Contributed to **roadway drainage and culvert design** through engineering calculations, quantity/cost estimates, and project documentation
- Prepared roadway and bridge drawings using **Civil 3D/AutoCAD**; supported technical specifications and bid documents
- Supported construction inspections and **QA/QC on 12+ road and bridge projects**

Research Assistant – Center for Water Resources, Nepal Aug 2022 – Jan 2023

- Applied Pfafstetter watershed classification and river-network coding across **6,000+ rivers** with national agencies
- Contributed to a national hydropower assessment examining water-resource demand and climate risks

SELECTED MODELING PROJECTS

- **Mid-Atlantic U.S.:** Validated simulated river discharge and water-level behavior against observations; assessed storm-related flooding using CaMa-Flood
- **Great Lakes Basin:** Integrated river discharge, water withdrawals, irrigation, dams, and infrastructure datasets in a GIS-based water-resources assessment
- **Red Cedar River, Michigan:** Conducted 2D hydraulic modeling using **SRH-2D/SMS**
- **Kalamazoo Watershed, Michigan:** Surface–groundwater modeling using **SWAT/MODFLOW**
- **Bagmati River, Nepal:** Hydrologic and hydraulic modeling using **HEC-HMS/HEC-RAS**

PUBLICATIONS & PRESENTATIONS

- Coauthored **five peer-reviewed journal articles**, including first-author papers in *Journal of Hydrology* (2025; Asian Water Tower river systems) and *Heliyon* (2024; Nepal hydropower)
- Selected for an **AGU Fall Meeting 2026 oral presentation** on assimilating NASA SWOT water-surface elevation to constrain river and reservoir states

HONORS & ADDITIONAL CREDENTIALS

- NSF-funded WaterCube NRT Fellow (2024–2025); merit-based graduate scholarship, MSU
- Registered Engineer, Nepal Engineering Council (2022); Autodesk Civil 3D and AutoCAD course certificates (2020)